

Lakshmi Bharathy Kumar

San Jose,

CA | 650-275-4940 | lakshmibharathyk@gmail.com | linkedin.com/in/lakshmi-bharathy-kumar | github.com/Lakshmibharathy11

Authorized to work in the U.S. without sponsorship

SUMMARY

Data Scientist with an M.S. in Applied Data Intelligence from SJSU (May 2026) and 2 years of experience in data-driven marketing analytics. Focused on predictive modeling, statistical analysis, end-to-end data pipelines, and business decision support.

TECHNICAL SKILLS

Languages: Python, SQL

Statistics & Modeling: Regression, Classification, A/B testing, hypothesis testing, EDA, feature engineering, model evaluation, SHAP

ML & Deep Learning: scikit-learn, PyTorch, XGBoost, LSTMs, TF-IDF, MobileNetV2, Statsmodels

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, DAX

Data Engineering & Cloud: Airflow, dbt, Kafka, Snowflake, PostgreSQL, AWS S3, Docker

PROJECTS

Price Elasticity Prediction for SKU Pricing Strategy

Python, Pandas, Statsmodels, Power BI

- Analyzed 131,706 retail transactions; filtered 5,242 SKUs down to 107 with meaningful price variation and built log-log regression models to estimate price elasticity of demand per SKU.
- Identified 10 statistically significant SKUs ($p\text{-value} < 0.05$, $R^2 > 0.40$); top SKU EGB8E showed elasticity -1.39 at $R^2 = 0.596$ — directly actionable for pricing and promotion strategy.

Tesla Market Predictor

Airflow, Snowflake, dbt, Python, Sentiment Analysis

- Built an end-to-end pipeline (Airflow, Snowflake, dbt) combining Tesla stock data with 50,000+ social media sentiment records; engineered sentiment-weighted features and trained a regression model achieving $R^2 = 0.91$.

Developer Salary Prediction & Technology Trend Analysis

Python, Pandas, scikit-learn, Power BI

- Cleaned 2020–2024 Stack Overflow Survey data (49.14% missing salary); standardized experience and education fields, removed IQR outliers; prediction model reached $R^2 = 0.6241$, RMSE \$31K; Power BI dashboard surfaced compensation trends by role and region.

Customer Lifetime Value Segmentation & Churn Risk

Python, Pandas, scikit-learn, XGBoost, SHAP, Power BI

- Joined 9 Olist e-commerce tables (100K+ orders, 96K customers) to engineer an RFM feature matrix; K-Means clustering ($k=4$, silhouette 0.58) revealed the top 18% of customers by monetary score drove 61% of total revenue across all three years of transaction data.
- XGBoost churn classifier (ROC-AUC 0.84, F1 0.79) on 12-month behavioural features; SHAP identified recency and inter-purchase gap as the two strongest predictors; CLV regression ($R^2 = 0.73$) showed At-Risk customers had 3.2x higher predicted 6-month value than Lost customers.

Corporate Bankruptcy Early Warning System

Python, Pandas, XGBoost, SHAP, Statsmodels

- Engineered 47 financial ratios from income statement, balance sheet, and cash flow data across 3,722 companies; applied temporal cross-validation (train years 1–3, predict years 4–5) to enforce a strict 2-year prediction horizon and prevent data leakage.
- Benchmarked Altman Z-Score (ROC-AUC 0.71) against XGBoost (ROC-AUC 0.84); achieved 76% recall on true bankruptcies at 14% FPR; SHAP flagged debt-to-equity trajectory and cash flow margin as the two strongest early-warning signals.

PUBLICATIONS

L. B. Kumar, G. Liu, and Y. Luo, “Comparative Evaluation of Eye-Tracking Feature Representations for Gaze Direction Classification in Assistive Human–Machine Interaction,” *manuscript in preparation*, 2026.

EDUCATION

M.S., Applied Data Intelligence

May 2026

San Jose State University, San Jose, CA

- Thesis* — Evaluated 5 eye-tracking feature types and 7 model architectures (5 ML + 2 DL) for 5-class gaze classification; applied GridSearchCV tuning, stratified cross-validation, and per-class F1 reporting — RF hit 0.90 (user-dependent), SVM/GBDT reached 0.94 cross-user, 12 pp above the published baseline.

B.E., Electronics and Communication Engineering

2021

Anna University, Chennai, India

EXPERIENCE

Digital Marketing Specialist — RepuNEXT, Chennai, India

Jun 2021 – Dec 2022

- Analyzed website traffic and campaign performance using Google Analytics and SEMrush to isolate drop-off points; updated SEO targeting based on findings, growing organic traffic 25%.
- Built automated reporting for weekly campaign metrics and audience segmentation; ROI improved 15%.